Seaways MPSS

Marketing landing page for **Seaways MPSS** — LANG's **Multi-Purpose Semi-Submersible**, a validated, standardized deepwater production host built on a five-S design philosophy: **Simple · Safe · Swift · Size · Saves.** The site positions the MPSS as pre-positioned, leaseable offshore infrastructure that compresses the timeline from FID to first production.

Structure

FILE	PURPOSE
`index.html`	The full single-page site (hero, the MPSS, the Five S, specs, the model, heritage, leasing, contact).
`day-rates.html` + `day-rates.js`	An editable day-rate leasing model — swap tenants in/out, compare revenue, net and payback; export to CSV.
`invest.html` + `invest.js`	**For Family Offices.** An investor page framing the MPSS as a hard, income-producing real asset, with an interactive equity-returns calculator: type in a check size and see ownership, cash yield, IRR, MOIC and payback compute live, plus a year-by-year distribution schedule and CSV export.
`memo.html` + `memo.css`	**Founding Investor Memorandum.** A brochure-style, print-friendly memorandum: cover page, numbered sections (opportunity, problem, solution, market, business model, illustrative economics, heritage, founding round & use of proceeds, the ask) and a full disclaimer. Reuses `styles.css`; `memo.css` adds the brochure layer.
`styles.css`	All styling. Navy / ocean-blue / steel palette; fully responsive.
`main.js`	Mobile nav toggle and footer year. The site works without JS (the calculators need JS).

Running locally

It's a static site — open `index.html` in a browser, or serve the folder:

```bash python3 -m http.server 8000
```

then visit http://localhost:8000

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## Deploying

Any static host works (GitHub Pages, Netlify, Cloudflare Pages, S3, etc.). For the `seaways-mpss.com` domain, point the host at this repository / folder.

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## Content notes

- **Contact details** — search `index.html` (and `invest.html`) for `stewart.lang@seaways-mpss.com` and update the email; add phone/address in the `#contact` / `#data-room` sections as needed.
- **Investor figures** are illustrative placeholders defined in `defaultState()` in `invest.js` (asset cost, leverage, interest, net charter cash flow, hold, residual). Edit there to change the defaults; every field is also editable live in the browser. The page carries a clear disclaimer that the model is for discussion only and is not an offer of securities.
- **Copy** lives inline in `index.html`, grouped by clearly-commented sections.
- **Founding-round amount** — the memorandum's "The Ask" shows a placeholder `$[__]M` (marked `data-illustrative` in `memo.html`). Replace it with the real figure before sharing. The use-of-proceeds percentages and milestone phases in the same section are illustrative too.
- Specs, the Five S and the lifecycle/driver tables are hand-built in HTML/CSS (no third-party chart images), so they're safe to edit and reuse.
- Key figures (15,000 t payload, 8,100 m² deck, 2M bbl SWIS storage, 37.5 m survival wave, ~14-month hull delivery, LR/ABS/DNV approval) and the design heritage are drawn from Seaways Engineering source material.

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SWOT — Renewable projects · Support vessels · The MPSS

Internal working analysis · August 2026 · Not investor-facing copy. Nothing below should be pasted into `index.html`, `memo.html` or `invest.html` without rewriting — it is deliberately critical of the current positioning.

Three SWOTs and a comparison, prepared to test one question the site and the memorandum currently leave open: **how much of the MPSS case should rest on renewables?**

Scope — three different businesses, not three versions of one

UNIT OF ANALYSIS	WHAT IT COVERS	POSITION
A. Renewable projects	Offshore wind, fixed-bottom and floating, plus the adjacent transition missions the site already sells against: CCS, offshore green hydrogen, offshore power and BESS, offshore data centres.	Asset owner / developer
B. Support vessels	WTIVs and offshore heavy-lift, SOV/CSOVs, CSVs and cable-lay, feeder barges, AHTS and tow spreads for floating wind.	Contractor / tonnage owner
C. The MPSS	A moored, low-motion 90 × 90 m box semi: 30,000 t deck payload, 8,100 m² deck, single 27 m operating draft, ~14-month hull, LR/ABS/DNV reviewed, leased on multi-year day rate.	Infrastructure landlord

That distinction drives the whole comparison.

A. SWOT — Renewable projects

Strengths *(internal · helpful)*

- **Long-dated contracted revenue.** CfDs, PPAs and offtake agreements run 15–20 years — longer than any oil-and-gas charter, and largely insulated from the commodity price.
- **Policy-manufactured demand.** Where the mandate holds, demand is legislated rather than discovered: net-zero targets, seabed leasing rounds, CCS storage licences.
- **Investment-grade counterparties.** Utilities and majors — Ørsted, Equinor, EnBW, RWE, bp — not marginal E&P independents.
- **Real technical adjacency for a low-motion host.** Floating wind, offshore hydrogen, CO_2 injection and offshore data all need what the MPSS sells: deck area, payload, station-keeping, low heave.

Weaknesses *(internal · harmful)*

- **The economics do not carry a $600 M host.** A semi-submersible floating-wind substructure runs roughly €430–485 k per MW installed — about €15 M for a 12 MW unit. One 15 MW machine on a corner column is ~€6–7 M of substructure value on a platform the memo prices at $600 M.
- **Power is the lowest-value cargo per m² of deck.** Wind sells electrons at a strike price; oil sells barrels. The gap shows up directly in the day rates below.
- **Chronic schedule slip.** Consenting, grid connection, port capacity and turbine supply each delay first power independently; the compounded schedule is the sector's defining risk.

- **Fabrication and quay capacity are the real constraint** on floating wind — not hull design. The MPSS does not relieve that bottleneck; it competes for the same quay.

Opportunities *(external · helpful)*

- **Floating wind is still pre-industrial** — no dominant substructure design, no serial production line, no incumbent. The MPSS's serial-build thesis aims at precisely this gap.
- **The adjacent missions are emptier than wind.** Offshore CCS, hydrogen and data centres have no standard host at all — a blanker sheet than wind.
- **Offshore data is the outlier**: power-hungry, latency-tolerant if sited well, backed by counterparties whose willingness to pay dwarfs anything in the wind supply chain.
- **A 2026 recovery is being called** after two brutal years, with UK AR7 adding secured floating offtake.

Threats *(external · harmful)*

- **A live wave of cancellations.** EnBW halted its 3 GW Mona and Morgan projects in the Irish Sea; the 58.4 MW Blyth 2 floating project was cancelled in January 2026; in February 2026 the US suspended leases on five large projects including Empire Wind.
- **Strict capital discipline has replaced growth at any cost.** Every developer is cutting scope — not adding a novel, unproven host to the critical path.
- **A thin floating pipeline.** 2026 added roughly 193 MW of secured floating offtake — a fraction of one MPSS's worth of capital.
- **Political reversibility.** Unlike an oil field, the demand is a policy artefact and can be withdrawn by an election, as the US lease suspensions show.

B. SWOT — Renewable-support vessels

Strengths *(internal · helpful)*

- **A proven, liquid business model.** Day-rate chartering is a century-old bankable structure with an established owner base, standard charterparties and deep second-hand markets.
- **Very high rates at the top of the fleet.** High-spec WTIVs are valued around $500 k/day, with wait-on-weather for a full spread at $18–22 k/hour in 2026.
- **Mobility is the core asset.** When the North Sea softens the hull sails to Taiwan or the US East Coast — the single most valuable structural feature the MPSS does not have.
- **O&M demand is structural.** Every turbine installed becomes 20–25 years of SOV demand whether or not new FIDs are taken.

Weaknesses *(internal · harmful)*

- **Utilisation risk sits with the owner.** Idle steel earns nothing, and the spread between a chartered and an idle year is the whole return.
- **Long charters earn thin returns.** Guaranteed-utilisation SOV charters trade at comparatively low day rates and often yield single-digit IRRs — the safety is priced in.
- **Turbine upscaling obsoletes tonnage.** A jack-up specified for 8 MW machines cannot install 15 MW. Modern SOVs earn €45–60 k/day, ~30% above 2020 — but that is a rate for the current generation only.
- **Capital-intensive with no alternative use.** A WTIV outside offshore wind is a stranded asset with a very short buyer list.

Opportunities *(external · helpful)*

- **Europe tightens later in the decade.** Supply of ≥15 MW-capable WTIVs is forecast tight from 2032, and from 2030 in other markets. Patient, well-timed tonnage gets paid.
- **Floating wind re-shuffles the vessel mix**, shifting work from jack-ups toward tow, mooring and offshore-assembly spreads — an opening for new asset types.
- **Consolidation is under way.** Fleet sales signal a maturing sector where scale owners buy distressed tonnage cheaply.

Threats *(external · harmful)*

- **A structural oversupply, now arriving.** More than 60 WTIVs and heavy-lift vessels deliver between 2026 and 2030; ≥15 MW-capable supply outside China goes from none in 2020 to over 25 by 2028.
- **The same story in the service fleet**: around 60 SOV/CSOV deliveries by end-2026 against 40 in 2025, into a market widely forecast oversupplied through 2030.
- **Rate collapse is the base case, not the tail.** The 2022–24 record rates were a shortage artefact, and the orderbook is fixing the shortage.
- **Cancellations hit the vessel market twice** — less installation work now, and fewer turbines to service for the next 25 years.

C. SWOT — The MPSS

Strengths *(internal · helpful)*

- **The engineering risk is genuinely behind it.** Four decades of work, 1:100 tank testing, doctoral research at Glasgow and Newcastle, peer-reviewed publication, and classification review by Lloyd's Register, ABS and DNV. Very few offshore concepts can say this.
- **The build sequence is the real differentiator.** Deck box outfitted at quay level, hull floated under it last (`render/mpssdeckloadingscenes.md`, Scenes 2–6). That removes the afloat lift and ~25–30 m of hook height — not the crane, but the most expensive hour in the schedule.
- **Load path integrity.** Tower centreline 7.5 m inboard lands dead centre on a 15 m square column, putting point loads into the ring pontoon rather than a deck span.
- **Low motion by construction.** A deep ring pontoon at 27 m sees ~40% of surface wave height; tested to a 37.5 m survival wave with no water on deck and no change of draft.
- **Standardisation is the actual product.** Serial hulls, one classed design, topsides swapped per mission — leaseable inventory available *before* FID. Nobody in floating wind has industrialised anything comparable.

Weaknesses *(internal · harmful)*

- **The renewable tenants cannot pay the required rate.** On the repo's own inputs, the ~15% unlevered yield in `memo.html` is achievable on exactly one preset: oil production.
- **A naming error diligence will catch.** `day-rates.js:15` sells "Offshore wind turbine install" at $180 k/day. The MPSS is a moored host with no heavy-lift crane and no transit mode — it is not an installation vessel. The turbine in the renders stands *on* the platform; it is not installed *by* it.
- **Immobile by design.** The mobility that lets a WTIV chase demand across basins is exactly what a moored host gives up. Redeployment is a de-mob, tow and re-moor campaign, not a voyage.
- **Not yet built.** Classed and reviewed is not delivered. First-of-class risk, yard slot risk and the ~14-month hull claim are unproven in steel.
- **Single-asset concentration.** One $600 M hull, one tenant at a time; a vessel owner diversifies across a fleet.

Opportunities *(external · helpful)*

- **Reposition wind from generator to hub.** The value is not one turbine on a corner column — it is the substation, BESS, conversion, hydrogen and accommodation functions for a whole floating array, on one classed platform that also survives a 37.5 m wave.
- **Offshore data is the strongest transition tenant in the model below** — the only non-hydrocarbon preset that clears the assumed 8% debt cost with room to spare.
- **CCS and hydrogen sell term, not rate.** 12-year and 10-year charters at 8.2–8.3% unlevered are the closest renewable analogue to infrastructure yield — and appeal to a different investor than the memo addresses.
- **Counter-cyclical yard pricing.** Cancellations free yard slots and soften steel and fabrication pricing. The best time to order serial hulls is a bad market.
- **Vessel oversupply is an argument *for* the MPSS**: a permanently moored host substitutes for chartered tonnage over a 10–20 year life, converting volatile day-rate exposure into a fixed one.

Threats *(external · harmful)*

- **The hydrocarbon core carries the returns while the renewable story carries the marketing.** Any investor who runs the day-rate model finds this in ten minutes. It should be stated first, not discovered.
- **Renewables counterparties are cutting, not committing.** Under strict capital discipline, no developer adds a first-of-class host to the critical path.
- **The market will compare against known things** — a WindFloat-class floater, a chartered CSOV, an FPSO conversion — not the "conventional bespoke FPU" the site benchmarks against. That table answers a question nobody in renewables is asking.
- **Illustrative figures still anchor.** $600 M, ~15%, 2× MOIC and 60% residual are placeholders in `invest.js`, but they read as claims.
- **First-mover risk is unpriced.** Serial production needs a second and third order that do not exist until the first hull earns.

D. The comparison

D.1 Every tenant preset, run against the investor model

Each preset in `day-rates.js` at face value, against the `invest.js` deal — $600 M asset, 55% debt at 8%, so $26.4 M/yr interest on $270 M of equity:

TENANT PRESET	TERM	NET $M/YR	UNLEVERED YIELD	EQUITY CF	CASH-ON-EQUITY
Oil production host (FPSO)	8 y	94.0	**15.7%**	$67.6M	**25.0%**
Drilling support	3 y	72.9	12.2%	$46.5M	17.2%
Offshore data center	10 y	55.5	9.2%	$29.1M	10.8%
Green hydrogen	10 y	49.8	8.3%	$23.4M	8.7%
CCS / carbon storage	12 y	49.3	8.2%	$22.9M	8.5%
Power generation	10 y	37.8	6.3%	$11.4M	4.2%
Offshore wind turbine install	2 y	36.5	6.1%	$10.1M	3.7%
Accommodation / flotel	2 y	23.4	3.9%	−$3.0M	**−1.1%**

Three things fall out of this table:

1. **The memorandum's headline is the oil case.** `invest.js` defaults to `netCharter: 94000000` — which is, to the dollar, the oil-production preset at 92% utilisation. The "~15% illustrative unlevered net yield" in `memo.html` §06 is not a blended figure and not a renewable one.
2. **Leverage only works above 8%.** At the assumed debt cost, gearing is accretive for oil, drilling, data centre, hydrogen and — marginally — CCS. For power generation, wind install and accommodation the debt eats the spread, and the flotel case goes cash-negative.
3. **Term and rate trade in the right direction.** CCS at 12 years and hydrogen at 10 pay less per day but far more predictably than 2-year wind or flotel work. For an infrastructure buyer, 8.2% across twelve years is a *better* asset than 12.2% across three — an argument the memorandum does not yet make.

D.2 Where the three sit structurally

DIMENSION	RENEWABLE PROJECTS	SUPPORT VESSELS	MPSS
Position in the chain	Asset owner / developer	Contractor, tonnage supplier	Infrastructure landlord
Revenue	Power price × output	Day rate × utilisation	Day rate × utilisation
Contract length	15–20 y (CfD / PPA)	2–10 y, spot to term	2–12 y by mission
Who holds utilisation risk	Developer (wind resource)	**Owner**	**Owner**
Mobility	Fixed for life	**High — chases demand**	Low; moored, tow to redeploy
Current supply balance	Demand shrinking	**Oversupplied 2025–2030**	No supply — none built
Rate direction	Strike prices under pressure	Softening from 2022–24 peak	Unproven
Capital per unit	€430–485 k/MW substructure	~$300–500 M+ high-spec WTIV	$600 M/host (illustrative)
Obsolescence	25–30 y asset life	**High — turbine upscaling**	Low — topsides swap, hull stays
Residual value	Site-bound, decom liability	Liquid but cyclical	Hard steel, redeployable
Counterparty	Government / utility	Developer / EPC	Operator / utility / hyperscaler
Main risk	Policy and consenting	Utilisation and rate collapse	**First-of-class, single asset**
Best return in this model	6.1–9.2% unlevered	Single-digit IRR on term charters	15.7% unlevered — on oil

D.3 What the comparison actually says

The MPSS is structurally closer to a vessel than to a renewable project — and it inherits the vessel model's weakness, owner-held utilisation risk, without its strength, mobility. It compensates with two things a vessel does not have: **topside reconfigurability**, which defeats the turbine-upscaling obsolescence that dates a WTIV, and a **27 m fixed draft with a 37.5 m survival wave**, which lets it hold station through weather that sends a CSOV to port. Those two features are the honest core of the advantage over tonnage, and the site currently buries them under the five-S pitch.

Against renewable projects, the MPSS is not a competitor and should stop implying it is. It cannot beat a €15 M semi-submersible floater as a 15 MW generator platform, and it cannot beat a $500 k/day WTIV as an installer. What it can do — and what no floater or vessel does — is hold a large, powered, low-motion, permanently-manned deck on station for twenty years. That is the natural home for an array substation, BESS, hydrogen production, CO_2 compression and injection, a data centre, or all of them at once: the argument Scene 5 of the render series already makes visually and the copy does not make commercially.

The timing argument cuts both ways. Vessel oversupply and project cancellations both make renewables a poor *first* tenant — nobody is committing capital to a first-of-class host in this market. The same conditions make it an excellent *second* one: a hull ordered into a soft yard market and paid for by an oil-production charter is positioned to take transition work when floating wind industrialises later in the decade, exactly as European ≥15 MW vessel supply tightens from 2030–32.

E. What follows from it

1. **Lead the memorandum with the hydrocarbon case and name it as such.** The ~15% figure is the oil-production preset. Saying so up front converts a diligence liability into a credibility asset, and makes the transition tenants optionality on top of an oil-underwritten return — a stronger structure than a blended claim nobody can reproduce.
2. **Fix `day-rates.js:15`.** Rename "Offshore wind turbine install" to something the MPSS can actually do — "Floating wind hub: substation, BESS & O&M base" — and re-rate it on a 10-year term. The current line invites the question "where is your crane?"
3. **Promote the offshore data centre.** It is the best-yielding non-hydrocarbon tenant in the model above — 9.2% unlevered, 10.8% on equity, 10-year term at 95% utilisation — and it faces a counterparty class with a genuinely different willingness to pay.
4. **Sell CCS and hydrogen on term, not rate.** Twelve and ten years of contracted cash flow at ~8% unlevered is an infrastructure product for an infrastructure buyer — a different and larger pool of capital than the family offices `invest.html` currently targets.
5. **Add the comparison the market will actually make.** The hull table in `index.html` benchmarks against a conventional twin-pontoon FPU. Add a second table against a floating-wind semi and a chartered CSOV — the MPSS wins on deck, payload, station-keeping and mission life, and that is a fight worth picking publicly.
6. **Use Scene 8 as the wind story, not Scene 4.** Four identical hulls under four loadouts is the standardisation argument. A single turbine being erected on a corner column is a $600 M platform doing a €15 M job, and a wind specialist will say so.

Sources

- Vessel Sector Deep Dive: WTIVs — Offshore Engineer
- 2026 WTIV Charter Rate Forecast — Oitha Marine
- SOV/CSOV Through 2030: Oversupply and Market Volatility — Maritime Magazines
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- Service operation vessels — Guide to a Floating Offshore Wind Farm
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- 2026 Global floating status: pipeline remains slim — Aegir Insights
- EnBW 3 GW UK project cancellation — EnkiAI
- Modelling the installation of next generation floating offshore wind farms — ScienceDirect
- Floating substructure — Guide to a Floating Offshore Wind Farm

Market figures are third-party estimates gathered August 2026 and are indicative only. All MPSS economics are this repo's own illustrative placeholders — not projections, offers or guarantees of return.